

Linear search

Revise 360

Worksheet for the 360° experience: Linear search depot

Name: _____ Class: _____ Date: _____

How to use this sheet

1. Discuss or research the Do Now on slide 1. Complete the starter and key terminology below.
2. Open Linear search depot. Read each station panel and write the key facts in your own words. Complete its worksheet tasks before tapping the badge.
3. Answer the questions on screen. Your first answer counts. Use review mode to learn from mistakes.
4. Copy your score into the yellow box, then complete the exam practice from memory.

By the end of the lesson I will:

- Describe and trace linear search.
- Interpret a search code segment.
- Compare successful and unsuccessful searches.

Starter: before you put the headset on

Search [18, 4, 27, 9] from left to right for 27. Which values are checked?

.....

.....

Key terminology

Linear search

.....

.....

Index

.....

.....

The stations

Stations 1 and 2: turn right. Stations 3 and 4: turn around. Stations 5 and 6: turn left. Final challenge: look down.

1 Check one at a time

Turn right

Key facts in your own words:

Example to trace:

List: 18, 4, 27, 9

Target: 27

Checks: 18, 4, 27

Found at index 2

Challenge: How many target comparisons find 27?

Order a successful search for 27 in [18, 4, 27, 9].

Put these steps in order: Report found and stop / Compare 27 with 27 / Compare 4 with 27 / Compare 18 with 27

Search [8, 6, 4] for 8. Comparisons?

2 Unsorted lists

Turn right

Key facts in your own words:

Example to trace:

UNSORTED

42, 7, 19, 3

Linear search: works

Sorting first: not needed

Challenge: Why can linear search use an unsorted list?

Which list can linear search use?

Must linear search sort its input first?

3 Unsuccessful searches

Turn around

Key facts in your own words:

Example to trace:

List: 8, 2, 6, 4

Target: 7

Check all four items

Result: not found

Challenge: How many target comparisons are needed here?

Search [3, 8, 1] for 9. Target comparisons?

An absent target in six items needs how many target comparisons?

4 Read the code

Turn around

Key facts in your own words:

Example to trace:

```
i = 0; found = false
while i < n AND NOT found
  if data[i] == target then
    found = true
  endif; i = i + 1
```

Challenge: What does `data[i] == target` test?

Why include `i < n`?

What does `i = i + 1` do in a linear search?

5 Index versus value

Turn left

Key facts in your own words:

Example to trace:

```
Index: 0  1  2  3
Value: 6  2  9  4
Target 9: index 2
Checks needed: 3
```

Challenge: State the value, index and comparison count for 9.

Match the quantities for finding 9 in [6, 2, 9, 4].

Match each item to a description from the list below.

3 / 2 / 9

Item	Matching description
Value found	
Zero-based index	
Target comparisons	

In [6, 2, 9, 4], index 1 stores...

6 Best and worst cases

Turn left

Key facts in your own words:

Example to trace:

List length: 100

First item: 1 comparison

Last item: 100

Absent: 100

Challenge: Why does a longer list worsen the worst case?

For 20 items, which case needs 20 comparisons?

The best-case linear search finds the target...

★ Final challenge (look down)

Before you tap the star, complete these tasks on paper.

Search [13, 5, 21, 8, 3] for 8. What is its zero-based index?

.....

In the same search, how many target comparisons occur?

.....

Your score

Copy it from the score screen as soon as it appears.

_____ / 19

Key question

How would you find an item in a box with no useful ordering?

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.....

.....

Review and improve

Use review mode. Explain one correction and why the correct answer fits.

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Exam practice

Near the end of the lesson, close the experience and answer from memory.

a) Describe a linear search for an absent target. [3]

b) Trace a search for 6 in [9, 2, 4, 6, 1]. Give checked values and its zero-based index. [2]

c) Explain one reason to use linear search for a short unsorted list. [2]

How confident am I?

Objective	Not yet	Getting there	Confident
Describe and trace linear search.			
Interpret a search code segment.			
Compare successful and unsuccessful searches.			